

Statistical Physics

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22 de abril de 2026

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Capítulo 1

Solids

1.1. Solids with N atoms in 3D

$$T = \frac{1}{2}m \sum_{i=1}^{3N} \dot{x}_i^2 \quad ; \quad x_i : \text{positions of } N \text{ atoms at time } t$$

$$\hookrightarrow T = \frac{1}{2}m \sum_{i=1}^{3N} \dot{u}_i^2 \quad \text{taking } u_i(t) = (x_i - x_i^0)$$

$$V(\{x_i\}) = V(\{x_i^0\}) + \sum_{i=1}^{3N} \left(\frac{\partial V}{\partial x_i} \right)_{\{x_i^0\}} (x_i - x_i^0) + \frac{1}{2} \sum_{i,j=1}^{3N} \left(\frac{\partial^2 V}{\partial x_i \partial x_j} \right)_{\{x_i^0\}} (x_i - x_i^0)(x_j - x_j^0) + \dots$$

At the *equilibrium position*: $V(\{x_i^0\}) = 0$ and $\left(\frac{\partial V}{\partial x_i} \right)_{\{x_i^0\}} = 0$. For small oscillations, higher order terms are neglected.

$$H = \sum_{i=1}^{3N} \frac{1}{2} m \dot{u}_i^2 + \frac{1}{2} \sum_{i,j=1}^{3N} K_{ij} u_i u_j \quad (1.1)$$

where $K_{ij} = \left(\frac{\partial^2 V}{\partial x_i \partial x_j} \right)_{\{x_i^0\}} \rightarrow$ Microscopic physics of the solid (its interactions).

But $H \neq \sum H_i$ due to the potential. As our framework needs $H = \sum H_i$, we perform a transformation to decouple it.

1.1.1. Normal mode decomposition

$$u = \begin{pmatrix} u_1 \\ \vdots \\ u_{3N} \end{pmatrix}. \text{ Then } T = \frac{1}{2} m \dot{u}^T \dot{u} \text{ and } V = \frac{1}{2} u^T K u \text{ (Our original Hamiltonian).}$$

We search for an orthogonal transformation ($S^T S = I$) over $u = Sq$ such that $S^T K S = \Lambda$, where Λ is diagonal.

$$\begin{aligned}\dot{u}^T &= \dot{q}^T S^T \\ u &= Sq\end{aligned}$$

then $H = \frac{1}{2}m\dot{q}^T\dot{q} + \frac{1}{2}q^T\Lambda q \rightarrow H = \frac{1}{2}\sum_{i=1}^{3N}m\dot{q}_i^2 + \frac{1}{2}\sum_{i=1}^{3N}\lambda_i q_i^2$

Which can be expressed as $H = \sum H_i \rightarrow H$ becomes a sum of independent oscillators!!

$$H = \sum_{i=1}^{3N} \left[\frac{p_i^2}{2m} + \frac{1}{2}m\omega_i^2 q_i^2 \right] \text{ with } p_i = m\dot{q}_i \text{ and } \omega_i = \sqrt{\frac{\lambda_i}{m}} \quad (1.2)$$

λ_i is the eigenvalue of K .

A very complicated movement can be decomposed as a sum of normal modes.

Example 1: Eq. of motion for the displacements

$$H = \sum_{i=1}^{3N} \frac{p_i^2}{2m} + \frac{1}{2} \sum_{i,j=1}^{3N} K_{ij} u_i u_j \text{ with } p_i = m\dot{u}_i.$$

$$\dot{u}_i = \frac{\partial H}{\partial p_i} = \frac{p_i}{m} \quad \dot{p}_i = -\frac{\partial H}{\partial u_i} \rightarrow m\ddot{u}_i = -\sum_{j=1}^{3N} K_{ij} u_j \implies \ddot{u} = \frac{-K}{m} u$$

$$\text{Since } u = Sq: S\ddot{q} = -S\frac{\Lambda}{m}q \implies \ddot{q} = -\frac{\Lambda}{m}q \implies \ddot{q}_i + \frac{\lambda_i}{m}q_i = 0 \implies \ddot{q}_i + \omega_i^2 q_i = 0$$

Solution: $q_i(t) = a_i \cos(\omega_i t + \phi_i)$ $u_i(t) = (Sq)_i \rightarrow u_i(t) = \sum_{j=1}^{3N} S_{ij} a_j \cos(\omega_j t + \phi_j)$
 a_j and ϕ_j are fixed by initial conditions.

Example 2: 2 atoms (1D)

$$V(u_1, u_2) = \frac{1}{2}ku_1^2 + \frac{1}{2}c(u_1 - u_2)^2 + \frac{1}{2}ku_2^2$$

$$u(t)?$$

$$\text{Initial conditions: } u_1(0) = u_2(0) = \alpha, \dot{u}_1(0) = \dot{u}_2(0) = 0.$$

$$V = \frac{ku_1^2}{2} + \frac{c(u_1 - u_2)^2}{2} + \frac{ku_2^2}{2} \implies K = k \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \quad (\text{assuming } c = k)$$

$$|K - \lambda I| = 0 \implies \begin{cases} \lambda_1 = k \\ \lambda_2 = 3k \end{cases} \implies \Lambda = \begin{pmatrix} k & 0 \\ 0 & 3k \end{pmatrix}$$

$$\text{For } \lambda_1 = k \rightarrow s_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \text{ (In-phase mode)}$$

$$\text{For } \lambda_2 = 3k \rightarrow s_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \text{ (Out-of-phase mode)}$$

$$S = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}. \text{ We can check } SS^T = I.$$

Initial conditions: $u_1(0) = \alpha, u_2(0) = \alpha, \dot{u}_1(0) = 0, \dot{u}_2(0) = 0$

$$u_i(0) = \sum_j S_{ij} a_j \cos(\phi_j) \rightarrow \begin{cases} \frac{1}{\sqrt{2}} a_1 \cos(\phi_1) + \frac{1}{\sqrt{2}} a_2 \cos(\phi_2) = \alpha \\ \frac{1}{\sqrt{2}} a_1 \cos(\phi_1) - \frac{1}{\sqrt{2}} a_2 \cos(\phi_2) = \alpha \end{cases}$$

Summing both: $2a_1 \cos(\phi_1) = \alpha\sqrt{2}$. Since $\dot{u}(0) = 0 \rightarrow \phi_j = 0$. The amplitude associated with the second mode is zero, $a_2 = 0$. Then: $u_1(t) = \alpha \cos(\omega_1 t)$ and $u_2(t) = \alpha \cos(\omega_1 t)$. The oscillators are *in phase*.

If $u_1(0) = \alpha$ and $u_2(0) = -\alpha \rightarrow a_1 = 0, a_2 = \sqrt{2}\alpha$. Then: $u_1(t) = \alpha \cos(\omega_2 t)$ and $u_2(t) = -\alpha \cos(\omega_2 t)$. Oscillators are *out of phase*.

ANY VIBRATION CAN BE DECOMPOSED AS A SUM OF NORMAL MODES

Solid as classical independent harmonic oscillators (same mass)

$$H = \sum_{i=1}^{3N} H_i = \sum_{i=1}^{3N} \frac{p_i^2}{2m} + \frac{1}{2} m \omega^2 q_i^2 \text{ where } p_i = m \dot{q}_i.$$

$$Q(N, V, T) = (z_1)^{3N} \quad z_1 = \frac{1}{h} \int \int dp dq e^{-\beta(\frac{p^2}{2m} + \frac{1}{2} m \omega^2 q^2)} = \underbrace{\frac{1}{h} \int e^{-\frac{\beta p^2}{2m}} dp}_{\sqrt{2\pi m k T}} \underbrace{\int e^{-\frac{\beta m \omega^2 q^2}{2}} dq}_{\sqrt{\frac{2\pi k T}{m \omega^2}}} =$$

$$\frac{2\pi}{h\beta\omega} = \frac{kT}{h\omega}$$

$$\text{Then } Q = \left(\frac{kT}{h\omega}\right)^{3N} = (c\beta)^{-3N}$$

$$A(V, T, N) = -kT \ln Q = 3NkT \ln \left(\frac{h\omega}{kT}\right)$$

$P = -\left(\frac{\partial A}{\partial V}\right)_{T,N} = 0$ (Pressure vanishes since ω is independent of V in this model).

$$S = -\left(\frac{\partial A}{\partial T}\right)_{V,N} = 3Nk \ln \left(\frac{kT}{h\omega}\right) + 3Nk$$

$U = A + TS = 3NkT = 3N \cdot 2 \cdot \frac{1}{2} kT$ ($3N$ particles, each with 2 degrees of freedom: kinetic and potential. Each provides $1/2 kT$ according to equipartition theorem).

$$C_v = \frac{1}{n} \left(\frac{\partial U}{\partial T}\right)_v = 3k \quad \text{This is not } C_v \text{ per mol, it is per particle. Then } C_v^{(m)} = C_v \cdot N_A = 3kN_A = 3R$$

$$C_v^{(m)} \approx 3R \approx 6 \text{ cal/mol} \cdot \text{K}$$

Dulong-Petit law

$$C_v^{(m)} \begin{cases} \text{Pb} \sim 6,43 \\ \text{Al} \sim 6,03 \end{cases} \implies \text{Simple solids}$$

For diatomic gases has higher $C_v^{(m)}$
 $C_v \sim \frac{1}{N} \left(\frac{\partial U}{\partial T} \right)_v \rightarrow$ energy needed to increase T by 1.

(In diatomic atoms there are more ways to use the energy, then is required more energy
 $\rightarrow$ Need more C_v)

Given a graph, we should be able to know which has higher C_v :

$$U = \alpha T \implies \frac{\partial U}{\partial T} = \alpha = C_v$$

If $\alpha_1 > \alpha_2 \implies C_{v1} > C_{v2}$

But there is a contradiction between C_v predicted and measured. Then Einstein proposed to take account the quantum behavior.

See in RAW graphics

Independent Quantum Harmonic Oscillator

$$H = \sum_{j=1}^{3N} H_j \quad , \quad \epsilon_n = (n + 1/2)\hbar\omega \quad \text{for } n = 0, 1, 2, \dots$$

$$z_1 = \sum_{n=0}^{\infty} e^{-\beta(n+1/2)\hbar\omega} \rightarrow \sum x^n = \frac{1}{1-x} \rightarrow z_1 = \left[2 \sinh \left(\frac{\hbar\omega\beta}{2} \right) \right]^{-1}$$

$$A = -kT \ln z_{3N} = 3N \left[\frac{1}{2}\hbar\omega + kT \ln (1 - e^{-\beta\hbar\omega}) \right]$$

$$z_{3N} = (z_1)^{3N}$$

$$P = - \left(\frac{\partial A}{\partial V} \right)_{T,N} = 0$$

$$S = - \left(\frac{\partial A}{\partial T} \right)_{V,N} = 3Nk \left[\frac{\beta\hbar\omega}{e^{\beta\hbar\omega} - 1} - \ln (1 - e^{-\beta\hbar\omega}) \right] \quad \text{Notice that when } T \rightarrow 0 \implies S \rightarrow 0$$

(Satisfying the Third Principle of Thermodynamics).

When $T \rightarrow \infty$ (or $\beta \rightarrow 0$):
 $e^x - 1 \approx x$ when $x \ll 1 \implies S = 3Nk \left[1 + \ln \left(\frac{kT}{\hbar\omega} \right) \right] \rightarrow$ Same result as we got for the classical model.

$$U = A + TS = 3N \left[\frac{1}{2}\hbar\omega + \frac{\hbar\omega}{e^{\beta\hbar\omega} - 1} \right]$$

$$\lim_{T \rightarrow 0} U(T) = \frac{3N\hbar\omega}{2} \quad (\text{Zero-point energy})$$

$$kT \gg \hbar\omega \implies e^{\beta\hbar\omega} - 1 \approx \beta\hbar\omega \implies U(t) \approx 3N \left(\frac{1}{2}\hbar\omega + kT \right)$$

Plotting this: *Classical result:* $U = 3NkT$ (*Equipartition theorem*). *Quantum result:* Approaches $3/2N\hbar\omega$ as $T \rightarrow 0$.

$$\text{Now we can determine } C_v: C_v = \frac{1}{N} \left(\frac{\partial U}{\partial T} \right)_v = \frac{3k(\beta\hbar\omega)^2 e^{\beta\hbar\omega}}{(e^{\beta\hbar\omega} - 1)^2}$$

$$\begin{cases} T \rightarrow \infty & C_v \approx 3k \\ T \rightarrow 0 & C_v \approx 3k \frac{(\beta \hbar \omega)^2 e^{\beta \hbar \omega}}{(e^{\beta \hbar \omega} - 1)^2} = \frac{(kT \hbar \omega)^2}{e^{\beta \hbar \omega}} \rightarrow 0 \end{cases}$$

Plotting C_v : Graph shows C_v saturating at $3k$ (Dulong-Petit) at high T and dropping exponentially as $T \rightarrow 0$.

Einstein Function: $\mathcal{E}(x) = \frac{x^2 e^x}{(e^x - 1)^2}$ $\Theta_E = \frac{\hbar \omega}{k} \rightarrow$ Einstein Temperature. We can express then: $C_v = 3k \mathcal{E}\left(\frac{\Theta_E}{T}\right)$

But experimental points did not fit exactly as the quantum model: Einstein model drops too fast (exponentially). Experiment shows a T^3 behavior at low temperatures.

Einstein model had assumed that particles are independent and have the SAME FREQUENCY.

Debye Model

$$H = \sum_{j=1}^{3N} H_j \quad , \quad H_j = \frac{p_j^2}{2m} + \frac{1}{2} m \omega_j^2 q_j^2 \quad , \quad Z = \prod_{j=1}^{3N} z_j$$

$$z_j = \sum_{n=0}^{\infty} e^{-\beta(n+1/2)\hbar\omega_j} = \frac{e^{-\beta\hbar\omega_j/2}}{1 - e^{-\beta\hbar\omega_j}}$$

$$A = -kT \ln Z = -kT \ln \prod z_j = \sum_{j=1}^{3N} \left[\frac{1}{2} \hbar \omega_j + kT \ln(1 - e^{-\beta\hbar\omega_j}) \right]$$

$$U = \sum_{j=1}^{3N} \left[\frac{1}{2} \hbar \omega_j + \frac{\hbar \omega_j}{e^{\beta\hbar\omega_j} - 1} \right]$$

$$C_v = \frac{1}{N} \left(\frac{\partial U}{\partial T} \right)_v = \frac{1}{N} k \sum_{j=1}^{3N} \mathcal{E}(\beta \hbar \omega_j) \quad \text{where } \mathcal{E}(x) = \frac{x^2 e^x}{(e^x - 1)^2}$$

Using limits: $\int f(\omega) g(\omega) d\omega$

$$\sum f(\omega_j) = \sum_j \int f(\omega) \delta(\omega - \omega_j) d\omega = \int f(\omega) \underbrace{\sum_j \delta(\omega - \omega_j)}_{g(\omega)} d\omega$$

$$\text{Then: } \sum_j f(\omega_j) = \int f(\omega) g(\omega) d\omega$$

I assume that $g(\omega)$ is a smooth function because I have a high number of modes, and I assume a *fixed medium*. I suppose that C_v can be written as:

$$C_v = \frac{1}{N} k \int \mathcal{E}(\beta \hbar \omega) g(\omega) d\omega$$

$g(\omega) \equiv$ density of modes.

$$\int_0^{\omega_D} g(\omega) d\omega = 3N \quad (\text{Total number of modes})$$

$\omega_D =$ cutoff or maximum frequency $\rightarrow$ Debye Frequency.

$$C_v = \frac{1}{N} k \int_0^{\omega_D} \mathcal{E}(\beta \hbar \omega) g(\omega) d\omega$$

The wavelength is the distance between two consecutive maximums or minimums. $n = 1 \rightarrow \lambda = 2L$, $n = 2 \rightarrow \lambda = L$, $n = 3 \rightarrow \lambda = 2L/3...$ As we grow in n , a minimum wavelength corresponds to $\lambda = 2a$ (a is lattice constant). Then: $\frac{\omega}{k} = c \implies \omega = ck_c = c \frac{2\pi}{\lambda_c} = c \frac{2\pi}{2a} = c \frac{\pi}{a} \rightarrow$ maximum frequency.

$\int g(\omega) d\omega \rightarrow$ gives us the total number of modes in a range. In this case, $g(\omega) \propto \omega^2$ for a 3D elastic medium. $g(\omega) = \frac{V\omega^2}{2\pi^2c^3}$ where c is the speed of sound.

$$\int_0^{\omega_D} g(\omega) d\omega = 3N \implies \int_0^{\omega_D} \frac{V\omega^2}{2\pi^2c^3} d\omega = \frac{V\omega_D^3}{6\pi^2c^3} = 3N \implies \omega_D = c \left(\frac{18\pi^2N}{V} \right)^{1/3}$$

$$C_v = \frac{3k}{N} \int_0^{\omega_D} \mathcal{E}(\beta\hbar\omega) \frac{V\omega^2}{2\pi^2c^3} d\omega = \frac{9k}{\omega_D^3} \int_0^{\omega_D} \frac{(\beta\hbar\omega)^2 e^{\beta\hbar\omega}}{(e^{\beta\hbar\omega}-1)^2} \omega^2 d\omega$$

Let $x = \beta\hbar\omega \implies dx = \beta\hbar d\omega$.
Using the Debye temperature $\Theta_D = \frac{\hbar\omega_D}{k}$:

$$C_v = 9k \left(\frac{T}{\Theta_D} \right)^3 \int_0^{\Theta_D/T} \frac{x^4 e^x}{(e^x-1)^2} dx$$

Case 1: $T \gg \Theta_D$ (High T)
 x is very small, so $\frac{x^4 e^x}{(e^x-1)^2} \approx \frac{x^4}{x^2} = x^2$.

Then $C_v \approx 9k \left(\frac{T}{\Theta_D} \right)^3 \int_0^{\Theta_D/T} x^2 dx = 9k \left(\frac{T}{\Theta_D} \right)^3 \frac{1}{3} \left(\frac{\Theta_D}{T} \right)^3 = 3k \rightarrow$ This is the classical result.

Case 2: $T \ll \Theta_D$ (Low T)
The integral limit goes to ∞ . $\int_0^\infty \frac{x^4 e^x}{(e^x-1)^2} dx = \frac{4\pi^4}{15}$.

Then $C_v \approx 9k \left(\frac{T}{\Theta_D} \right)^3 \frac{4\pi^4}{15} = \alpha T^3 \rightarrow$ **cubic dependence that we were looking for.**

- Experimental
- Theoretical (Debye Model)